OCS 7001 - Final Project Proposal: How does Bedload Sediment Fraction Affect Delta Channel Network Structure?

Maya Maes-Johnson

Abstract

PyDeltaRCM is a reduced-complexity numerical river delta model. Landlab is a suite of modeling tools that houses the ModelGrid class which is capable of producing of a network from input topography. This project will test the effect of initial bedload sediment fraction on the channel network structure of deltas using PyDeltaRCM for delta simulation and Landlab's NetworkModelGrid for network generation. Using a graph-theoretic approach, these networks can be analyzed to quantify changes in structure to test whether bedload sediment fraction significantly impacts network structure.

1 Introduction

River deltas play a significant role in sustaining critical habitat and contributing to a global economy. However, deltas face a variety of challenges including subsidence, sea level rise, and saltwater intrusion. In the wake of these challenges, sediment and freshwater diversion has been proposed as a solution to combat coastal land loss (Peyronnin et al. 2013), but the success of such projects requires an understanding of how sediment input impacts delta development.

Edmonds and Slingerland 2010 used numerical modeling to demonstrate that the cohesiveness of sediment controls the overall morphology of river deltas. Based on these findings, we expect that varying the bedload fraction of sediment (essentially, the fraction sand in PyDeltaRCM) should also significantly impact network structure.

A graph-theoretic approach treats the river delta as a network of nodes (inlets, outlets, bifurcations, and confluences) and links (channels). This allows for quantitative analysis of delta development. Tejedor et al. 2015 presents a summary of simple topologic characteristics for channel networks. This project will test the effect of different initial bedload sand inputs, using a graph theoretic approach to analyze significant changes in network structure.

2 Methods

2.1 PyDeltaRCM Model Setup and Parameters

PyDeltaRCM (Moodie et al. 2021) is the Python-based descendant of DeltaRCM (Liang, Voller, and Paola 2015) that uses probabilistic random walks to model deltas. For the purposes of this project, all parameters with the exception of bedload fraction of the input sediment (f_bedload) will remain as default. Four models with input sediment fractions of 0.25, 0.5, 0.75, and 1.0 will be run for 1250 timesteps (∼38 years) so as to allow for the reasonable development of channels from which networks can be extracted.

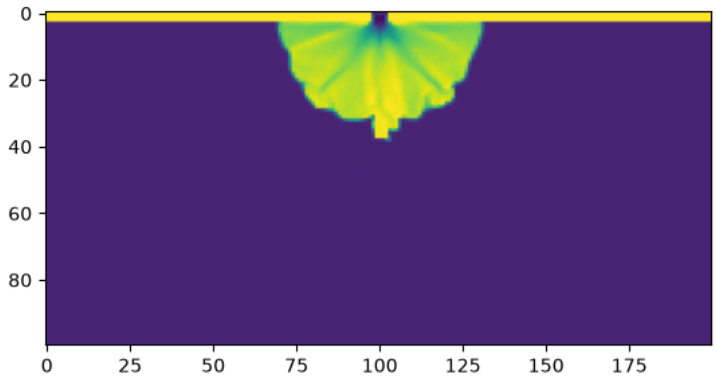


Figure 1: Output depth from a PyDeltaRCM model ran with default parameters for ∼ 10 years.

2.2 Network Extraction

Landlab is an open-source, community-developed library of tools for earth surface processes modeling (Hobley et al. 2017). Landlab supports sev-

eral forms of gridding including the network model grid. Using the network_grid_from_raster function from the NetworkModelGrid class, a network can be obtained from an ASC data file, and the number of nodes and links can be reduced to retain only the upstream and downstream nodes of each link (figure 2).

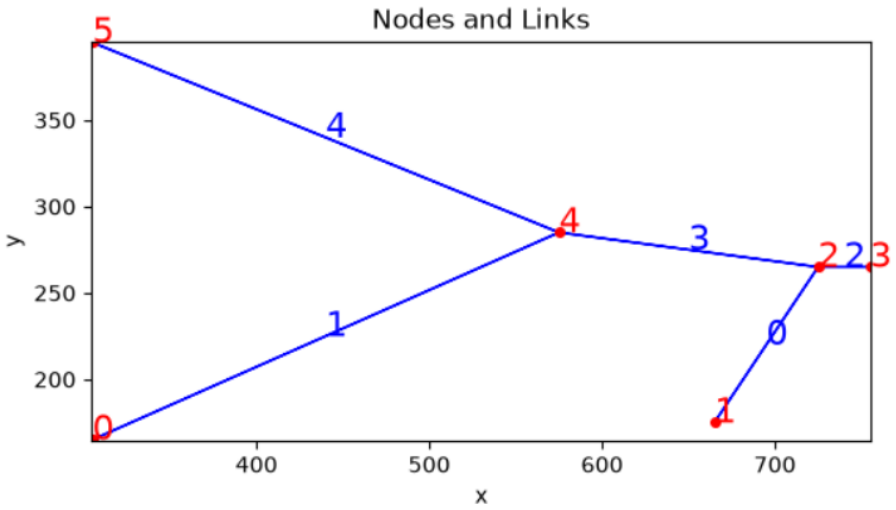


Figure 2: Reduced network model grid extracted from input topography.

Output from PyDeltaRCM will be transformed into an ASC file to obtain networks.

2.3 Network Analysis

Tejedor et al. 2015 provides a brief summary of simple topologic properties that can be derived from networks including number of links (N_L), number of vertices (N_V), and number of outlets (N_O). Using the reduced network model grids created from the PyDeltaRCM models, these properties can be manually extracted and compared.

3 Timeline

The following timeline is subject to change as the project progresses:

Timeline	
Week 1 (10/26 - 10/30)	Run PyDeltaRCM simulations
Week 2 (11/02 - 11/06)	Configure PyDeltaRCM output into ASC files
Week 3 (11/09 - 11/13)	Analyze network and visualize results
Week 4 (11/16 - 11/20)	Analyze network and visualize results, begin final report
Week 5 (11/23 - 11/24)	Write final report and format final presentation
Week 6 (11/29)	Make final edits

Table 1: Weekly task breakdown

4 Broader Impacts

This project uses two available, open-source products to develop a network analysis methodology that is efficient and easily replicable. The quantification of delta network structure has a relevant application in coastal restoration work as network analysis provides valuable information about delta self-organization. This can be useful in predicting the evolution of land-building deltas.

The Community Surface Dynamics Modeling System (CSDMS) which hosts the DeltaRCM and Landlab repositories encourages educational/ teaching material contributions. The end result of this project will be a first draft for a contribution to CSDMS.

References

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